

Name: _____ Date: _____

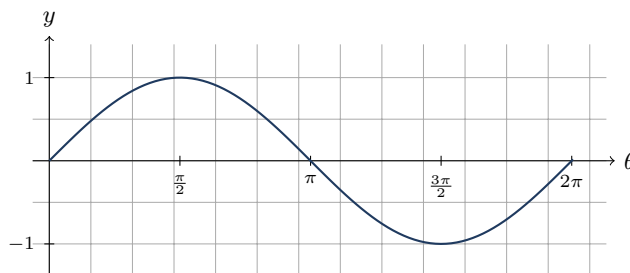
Period: _____

Spiral Review — Key

1. Evaluate each expression using your knowledge of the unit circle.

(a) $\sin\left(\frac{\pi}{2}\right) = 1$ (b) $\cos(\pi) = -1$ (c) $\sin\left(\frac{3\pi}{2}\right) = -1$ (d) $\cos(0) = 1$

2. The graph of $y = \sin \theta$ is shown below with key features labeled. Fill in each blank.



Amplitude = **1** Period = **2π** Midline: $y = 0$

3. A function f is transformed by a vertical stretch of factor 4 and a vertical shift up 2. If the parent function has the rule $f(\theta) = \sin \theta$, write the rule of the transformed function.

Transformed function: $f(\theta) = 4 \sin \theta + 2$

4. The function $g(\theta) = \sin(\theta - \pi)$ is a horizontal translation of $y = \sin \theta$.

- (a) In which direction is the graph shifted? **Right (positive direction)**
(b) By how many units? **π units to the right (phase shift = π)**